

Zhenghao “Mark” PENG

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LinkedIn Google Scholar

RESEARCH INTERESTS

Reinforcement Learning, Multimodal LLM (VLA, VLM) and Human-in-the-loop Learning.

EDUCATION

University of California, Los Angeles (UCLA)

September 2022 - March 2026

- Ph.D. in Computer Science, supervised by [Prof. Bolei Zhou](#).

The Chinese University of Hong Kong (CUHK)

August 2019 - July 2022

- M.Phil. in Information Engineering, supervised by [Prof. Bolei Zhou](#) at Multimedia Lab (MMLab).

Shanghai Jiao Tong University (SJTU)

Sept. 2015 - July 2019

- B.Eng. in Naval Architecture and Ocean Engineering. Member of Zhiyuan Honor Program.
- Research assistant supervised by [Prof. Li Jiang](#).

EXPERIENCE

Amazon, Sunnyvale, CA

April 2026 – Present

Member of Technical Staff at Amazon Frontier AI & Robotics (FAR)

- Foundation model post-training: Data curation, supervision design, and training recipe hillclimbing.
- Agentic navigation: Using robotics policies as tools for agents.
- Sim-to-real navigation-locomotion integration: System architecture and robustness to real-world variability. Achieved 100% object search success in an internal demo.

NVIDIA, Santa Clara, CA

June 2025 – December 2025

Research Intern in Autonomous Vehicle Group at NVIDIA Research

Manager: [Boris Ivanovic](#)

- Designed *Counterfactual VLA (CF-VLA)*, a self-reflective VLA framework that critiques and corrects its own actions before execution.
- Built a rollout-filter-label data engine that mines failure cases from model rollouts, filters scenes, and uses large VLM as a teacher to auto-label counterfactual reasoning traces.
- Integrated Qwen2.5-VL into the driving VLA with domain-specific video encoders/decoders, implemented multi-node FSDP training and batched rollouts, and automated dataset curation for 10M+ video clips.
- Experimented on large-scale driving datasets and showed consistent gains in all metrics, demonstrating up to 17% lower trajectory error, and 21% lower collisions than non-reasoning baselines. The model demonstrates **adaptive reasoning**: it thinks more in difficult, high-risk scenarios.

Waymo, Mountain View, CA

June 2023 - September 2023

Research Intern in Waymo Research

Mentored by [Justin Fu](#) and [Rowan McAllister](#)

- Fine-tuned multi-agent behavior models with reinforcement learning. Our method uses REINFORCE to train large transformer without the value net, similar to GRPO though much earlier than it.
- Reduced collision and off-road rates and improved safety-critical metrics on the Waymo Open Sim Agents Challenge (WOSAC), raising composite performance by 22%.
- Built a new evaluation framework that better ranks and tests autonomous driving planners in realistic scenarios generated by our model.

AWARDS AND HONORS

- Dissertation Year Award 2025-2026, UCLA
- Amazon Fellowship, 2024-2025, UCLA
- University Fellowship, 2023-2024, 2024-2025, UCLA

- The Outstanding Tutors Award 2021 of the Faculty of Engineering , 2021, CUHK
- Teaching Assistant Award, Term 1 2020 - 2021, Term 2 2020 - 2021, CUHK
- Postgraduate Studentship, 2019 - 2022, CUHK
- Zhiyuan Honors Scholarship, 2015 - 2017, SJTU

SELECTED PAPERS

For the complete list of publications, please visit my [Google Scholar](#). Selected papers:

- [13] **Alpamayo-R1: Bridging Reasoning and Action Prediction for Generalizable Autonomous Driving in the Long Tail.**
NVIDIA: Yan Wang, ..., [Zhenghao Peng](#), ..., Marco Pavone. ([arXiv:2511.00088](#), 2025) [[PDF](#)]
- [12] **Counterfactual VLA: Self-Reflective Vision-Language-Action Model with Adaptive Reasoning.**
[Zhenghao Peng](#), Wenhao Ding, ..., Marco Pavone. ([CVPR 2026 Highlight](#))
- [11] **Predictive Preference Learning from Human Interventions.**
Haoyuan Cai, [Zhenghao Peng](#), and Bolei Zhou ([NeurIPS 2025 Spotlight](#)) [[PDF](#), [Code](#), [Webpage](#)]
- [10] **Robot-Gated Interactive Imitation Learning with Adaptive Intervention Mechanism.**
Haoyuan Cai, [Zhenghao Peng](#), and Bolei Zhou ([ICML 2025](#)) [[PDF](#), [Code](#), [Webpage](#)]
- [9] **Data-Efficient Learning from Human Interventions for Mobile Robots.**
[Zhenghao Peng](#), Zhizheng Liu, and Bolei Zhou. ([ICRA 2025](#)) [[Webpage](#), [PDF](#)]
- [8] **Improving Agent Behaviors with RL Fine-tuning for Autonomous Driving.**
[Zhenghao Peng](#), Wenjie Luo, Yiren Lu, T. Shen, C. Gulino, A. Seff, and Justin Fu. ([ECCV 2024](#)) [[PDF](#)]
- [7] **Learning from Active Human Involvement through Proxy Value Propagation.**
[Zhenghao Peng](#), W. Mo, C. Duan, Q. Li, and Bolei Zhou. ([NeurIPS 2023 Spotlight](#)) [[PDF](#), [Webpage](#)]
- [6] **ScenarioNet: Open-Source Platform for Large-Scale Traffic Scenario Simulation and Modeling.** Quanyi Li*, [Zhenghao Peng](#)*, Lan Feng, Zhizheng Liu, Chenda Duan, Wenjie Mo, and Bolei Zhou. ([NeurIPS 2023](#)) [[PDF](#), [Code](#), [Webpage](#)]
- [5] **MetaDrive: Composing Diverse Driving Scenarios for Generalizable Reinforcement Learning.**
Quanyi Li*, [Zhenghao Peng](#)*, Z. Xue, Q. Zhang, and Bolei Zhou. ([TPAMI 2021](#)) [[PDF](#), [Code](#), [Webpage](#)]
- [4] **Efficient Learning of Safe Driving Policy via Human-AI Copilot Optimization.**
Quanyi Li*, [Zhenghao Peng](#)*, and Bolei Zhou. ([ICLR 2022](#)) [[PDF](#), [Code](#), [Webpage](#)]
- [3] **Safe Driving via Expert Guided Policy Optimization.**
[Zhenghao Peng](#)*, Quanyi Li*, C. Liu, and Bolei Zhou. ([CoRL 2021](#)) [[PDF](#), [Code](#), [Webpage](#), [Poster](#)]
- [2] **Learning to Simulate Self-Driven Particles System with Coordinated Policy Optimization.**
[Zhenghao Peng](#), Q. Li, K. Hui, C. Liu, and Bolei Zhou. ([NeurIPS 2021](#)) [[PDF](#), [Code](#), [Webpage](#), [Poster](#)]
- [1] **Non-local Policy Optimization via Diversity-regularized Collaborative Exploration.**
[Zhenghao Peng](#), Hao Sun, and Bolei Zhou. ([arXiv 2020](#)) [[PDF](#)]

OPEN-SOURCE PROJECTS

[MetaDrive](#) is a widely used open-source driving simulator for reinforcement learning and autonomous driving, with 1,200+ GitHub stars and 500+ citations (as of June 2026). For all my open-source projects, see [GitHub](#).

TALKS

- Human-in-the-loop Agent Learning, EECS 598: Action and Perception Guest Lecture, invited by: Stella Yu, May 2024

TEACHING EXPERIENCES

- CS260R Reinforcement Learning, UCLA, 2025 Winter & 2023 Fall & 2022 Fall
- IERG5350 Reinforcement Learning, CUHK, Term 1, 2021-22 & Term 1 and 2, 2020-21 & Term 2, 2019-20

MISCELLANEOUS

Review Service: NeurIPS, ICML, CVPR, ICCV, ICLR, CoRL, RSS, IROS, ICRA, AAAI, T-RO, RA-L, *etc.*

Skills: PyTorch, Jax, Ray, RLlib, ROS2, Docker, Transformers, FSDP, *etc.*